Research Statement

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This research statement presents some of my current and future projects. Overall, my research agenda aligns with the requirements of the position and would benefit stakeholders in Kentucky.

Current Projects

Project 1: Climate Response of Second-Season Corn in Brazil, with Marcelo Braga (UFV, Brazil). Job Market Paper

Second-season corn, known in Brazil as safrinha, is now the country's dominant corn crop, accounting for 80% of total corn production. It is planted after the main crop, usually soybeans, and grows during a drier period of the year. In this paper, we evaluate the effects of extreme heat and water deficit on safrinha yields using phenological windows (vegetative, flowering, early grain fill, late grain fill). We find that one degree day above 29°C reduces yields by 0.10%, with most of the damage coming from the vegetative and early grain-fill windows (see Figure 1). Water deficit is also damaging to yields. A 1°C increase in temperature is estimated to reduce yields by 8.2%.

Status: Submitted to *Agricultural Economics*.

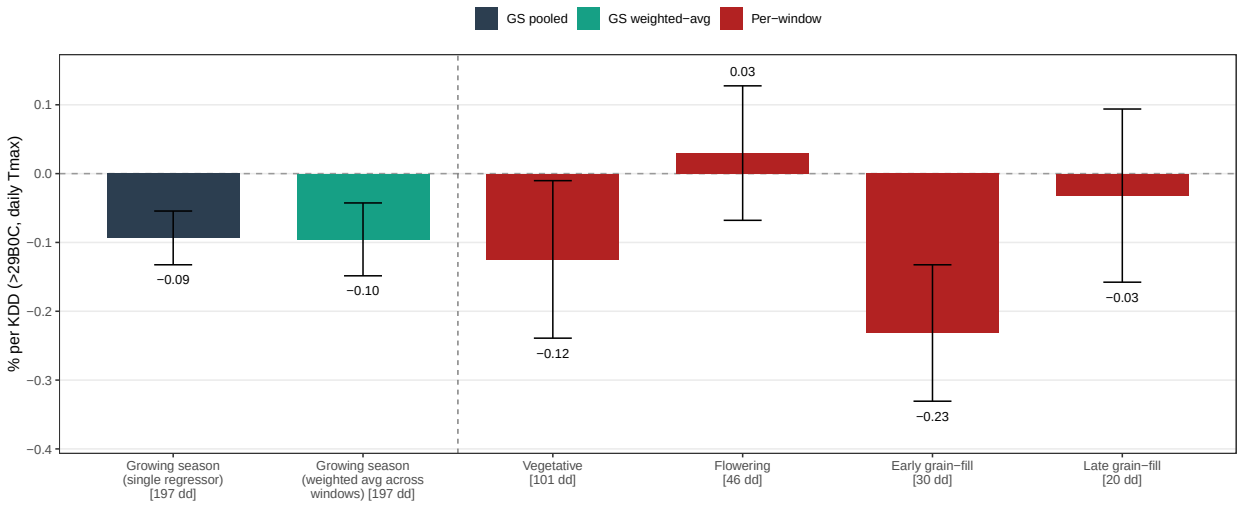


Figure 1: Extreme Heat Effects

Project 2: Dynamic Profit-Maximizing Fertilizer-Potassium Recommendations for Bermudagrass Hay (with Michael Popp, Nathan Slaton, and Gerson Drescher)

This is an interdisciplinary paper resulting from a research grant I worked on this summer with colleagues in Agricultural Economics and in Crop, Soil, and Environmental Sciences. In this paper, we use a seven-year fertilizer-K trial at two Arkansas locations and estimate time-varying bermudagrass yield, soil-test K (STK) transition, and K-removal equations to compare short-term, finite-horizon, and infinite-horizon profit-maximizing recommendations under constant and time-varying prices and costs. We find that yield exhibits diminishing marginal returns to total available K (see Table 1). Lagged STK and applied K significantly increase subsequent STK, whereas lagged yield is not statistically significant. The infinite-horizon approach maintains the highest terminal STK and a partial-return advantage over the short-term approach across the

seven-year horizon, with the difference increasing further when residual soil K is valued.

Status: Final edits before submission to JAAE.

Table 1: Estimates of the bermudagrass hay yield (Y) response to total available potassium (TAK) and K removal (KR) as a function of TAK and forage yield across three annual harvests.

Dependent variable	Y	KR
Constant	198.2510 (1080.0913)	-110.5912^{***} (11.0220)
TAK	-40.8241^{***} (7.4195)	-0.1350 (0.1005)
$\sqrt{\text{TAK}}$	854.7185^{***} (132.2252)	9.3164^{***} (1.5174)
TAK $\cdot\, t$	-5.3568^{***} (1.3788)	-0.0707^{**} (0.0226)
TAK $\cdot\, \sqrt{t}$	21.3445^{***} (5.3588)	0.1668 (0.0895)
Y	na	0.0146^{***} (0.0010)
Location × year FE	Yes	Yes
Replicate FE	Yes	Yes
Adjusted R^2	0.851	0.910
AIC	7155	3759
Observations	420	416

Standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Project 3: Irrigation and the Weather Sensitivity of U.S. Crop Insurance Losses

This paper examines how extreme heat and water deficit relate to federal crop insurance losses for corn and soybeans, and how that relationship differs between irrigated and non-irrigated production. Using county-level administrative data from the USDA Risk Management Agency for 2011–2023, restricted to Revenue Protection policies, we estimate how Extreme Degree Days above 29°C and growing-season water deficit relate to two outcomes: indemnity per acre and the loss cost ratio. For corn, the implied heat sensitivity of the loss cost ratio is about 99% smaller under irrigation, and the residual association is statistically indistinguishable from zero; for soybeans, it is about 81% smaller (see Figure 2). Water deficit follows a U-shaped relationship with losses, with both excess moisture and drought raising indemnities. Projected additional annual outlays for corn and soybeans rise to roughly $5.6 billion at +3°C, with non-irrigated production accounting for the overwhelming majority in every scenario.

Status: Final edits before submission to AEPP.

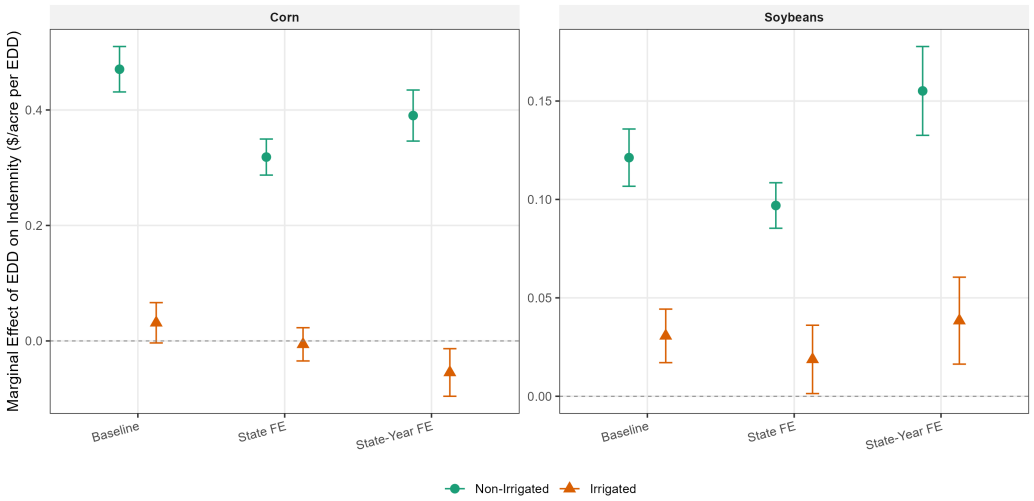


Figure 2: Extreme Degree Days Marginal Effects on Indemnity per Acre across Fixed Effects Specifications

Future Projects

Data Centers and the Impact on Agricultural Land Use

The impacts of data centers have been increasingly discussed recently. This is also a very hot topic in Kentucky (link). In this project, I aim to understand how data center entry affects surrounding farmland: does land exit agriculture? If so, what does it convert to? If not, do remaining farms change practices (crop choice, irrigation, etc.)?

Status: Data on data center entry timing and location already collected. Now working on merging this data with agricultural outcomes. Target: AJAE.

Financial Cooperatives and Agricultural Performance

Financial cooperatives are very popular in Brazil, and their share of total rural credit supplied has increased rapidly over the past two decades. In this project, my goal is to measure the impact of cooperative entry at the municipality level, using staggered difference-in-differences (Callaway and Sant'Anna) to exploit the staggered timing of entry. Preliminary results show a positive and significant impact of cooperative entry on crop revenue in treated municipalities (see Figure 3).

Status: Manuscript in preparation. Target: AJAE.

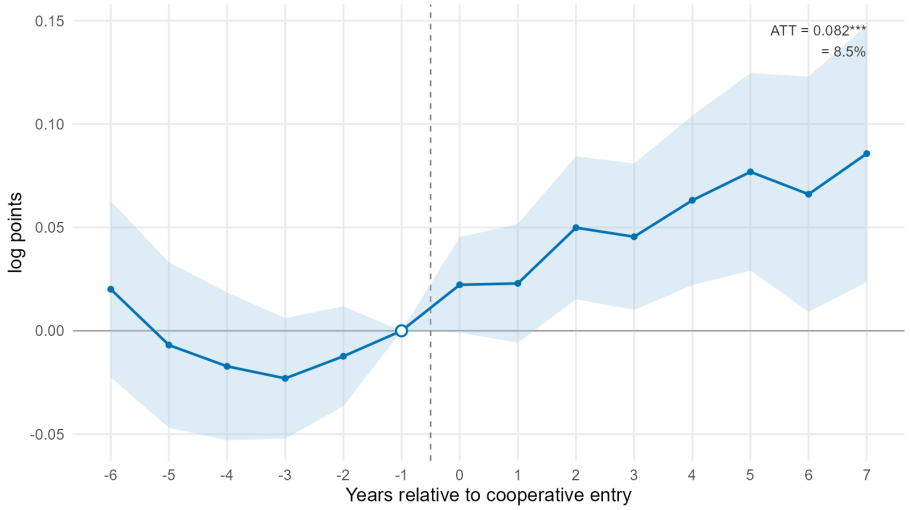


Figure 3: Effects of credit-cooperative entry on municipal crop value.

Conclusion

My research projects align very well with Kentucky's priorities, especially the impacts of data centers and of weather shocks on agricultural production. Given my experience working with faculty in the Crop, Soil, and Environmental Sciences department at the University of Arkansas, I want to build similar relationships at the University of Kentucky, both with colleagues in the department, and with researchers across the Martin-Gatton College of Agriculture, Food and Environment.